

Activity-6

Q1) i. **Why is the chart misleading?**

The chart is misleading because the Y-axis begins at 90 instead of 0. This exaggerates the visual differences between the bars. Although sales increased only from 95 lakhs to 105 lakhs (about 10.5%), the truncated axis makes June appear almost twice as large as January. This creates a false impression of dramatic business growth.

ii. **Visualization Principle Violated**

The violated principle is

➔ **Start the Y-axis at zero for bar charts**

Bar charts represent quantities using bar lengths. If the axis does not begin at zero, viewers perceive incorrect differences.

iii. **Recommended Visualization**

- Use a standard bar chart with the Y-axis starting at 0.

- A line chart is also suitable because it clearly shows monthly trends without exaggerating differences.

iv. **Business Impact**

This misleading chart may

- Convince management that sales growth is extraordinary.
- Lead to unnecessary increases in production.
- Cause incorrect budgeting.
- Influence investors to believe sales are increasing rapidly.

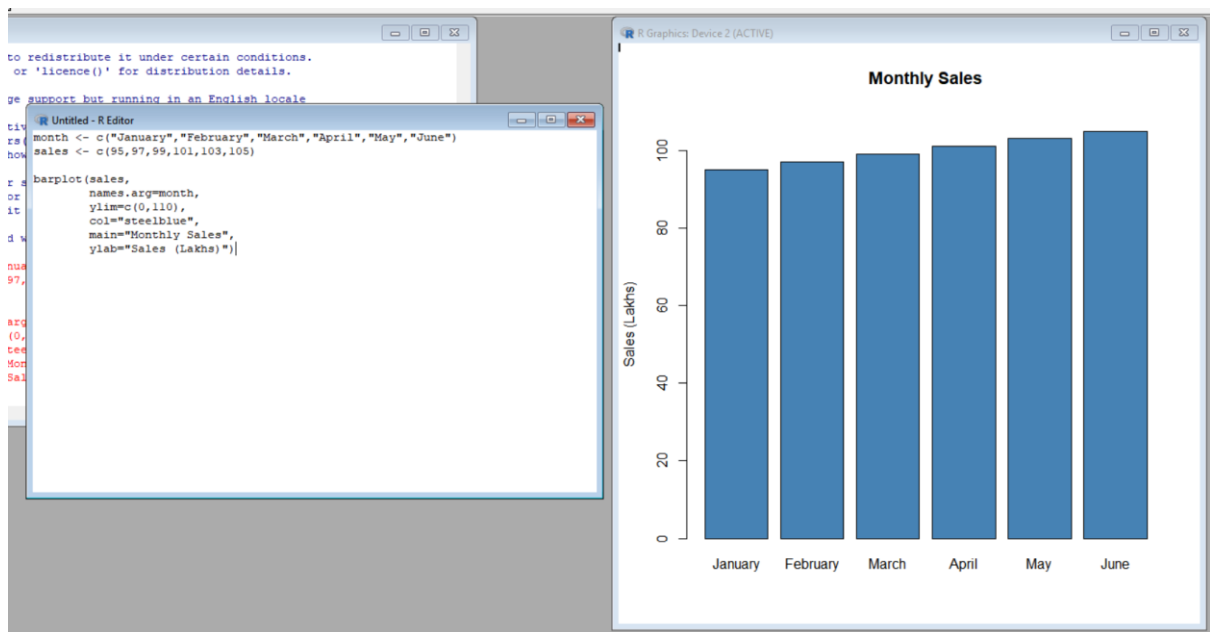
Therefore, decisions may be based on incorrect visual interpretation instead of actual data.

R code:

```
month <-  
c("January","February","March","April","May","June")  
sales <- c(95,97,99,101,103,105)  
barplot(sales,  
        names.arg=month,  
        ylim=c(0,110),
```

```
col="steelblue",  
main="Monthly Sales",  
ylab="Sales (Lakhs)")
```

Output:



Q2) i. **Correct Percentages**

Computer Science

$$420/1000 \times 100 = \mathbf{42\%}$$

Electronics

$$280/1000 \times 100 = \mathbf{28\%}$$

Mechanical

$$200/1000 \times 100 = \mathbf{20\%}$$

Civil

$$100/1000 \times 100 = \mathbf{10\%}$$

ii. **Why is it misleading?**

The displayed chart uses

- 50%
- 30%
- 15%
- 15%

instead of

- 42%
- 28%
- 20%
- 10%

The Civil department appears much larger than reality, while Mechanical appears smaller. Therefore, the pie chart does not represent the actual student distribution.

iii. **Better Visualization**

A bar chart is recommended because people compare lengths more accurately than pie slices.

iv. **Institutional Impact**

Incorrect enrollment visualization may

- Allocate more classrooms than required.
- Recruit unnecessary faculty.
- Purchase excess laboratory equipment.
- Mislead university administrators during planning.

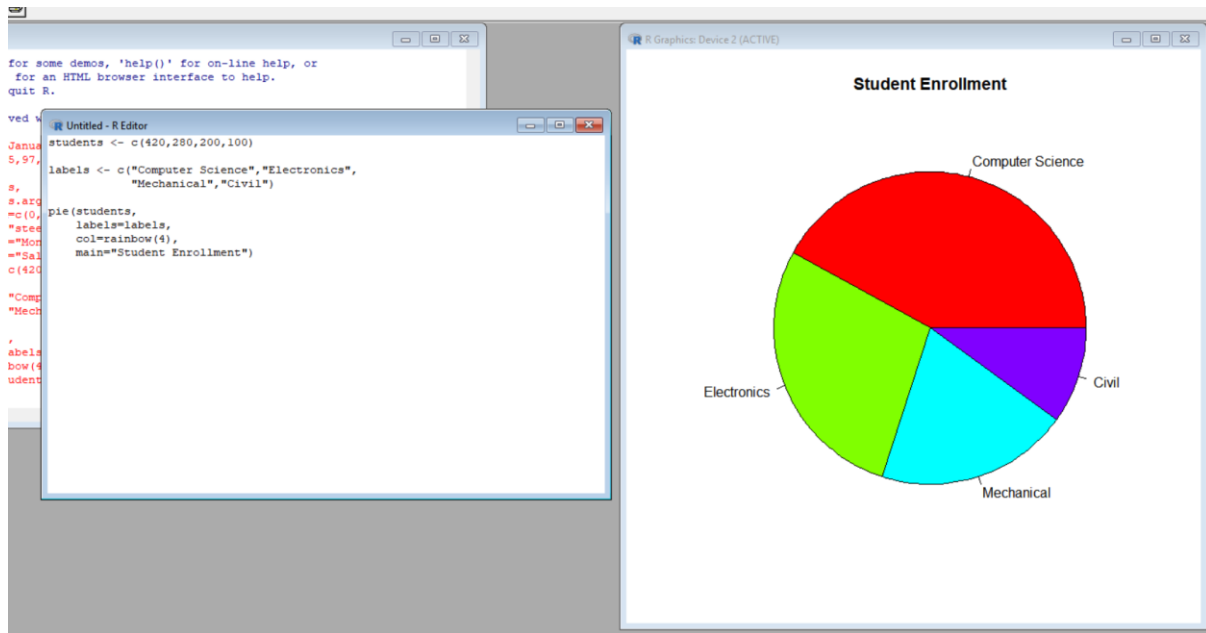
R code:

```
students <- c(420,280,200,100)
```

```
labels <- c("Computer Science","Electronics",  
            "Mechanical","Civil")
```

```
pie(students,  
    labels=labels,  
    col=rainbow(4),  
    main="Student Enrollment")
```

Output:



Q3) i. Graphical Error

The designer used **different bar widths**.

Only the heights should represent the values. Widths must remain equal.

ii. Why does it distort perception?

People observe the **area** of bars rather than only height.

A wider and taller bar appears much larger than its actual value.

Although Product D revenue is only

$105/60 = 1.75$ times

Product A,

the chart makes it appear nearly **three times larger**.

iii. **Correct Design**

Use

- Equal-width bars
- Same spacing
- Zero baseline
- Uniform color (or colors only for categorization)

iv. **Consequences**

Investors may

- Invest heavily in Product D.
- Ignore Product A.
- Overestimate company performance.
- Make poor investment decisions.

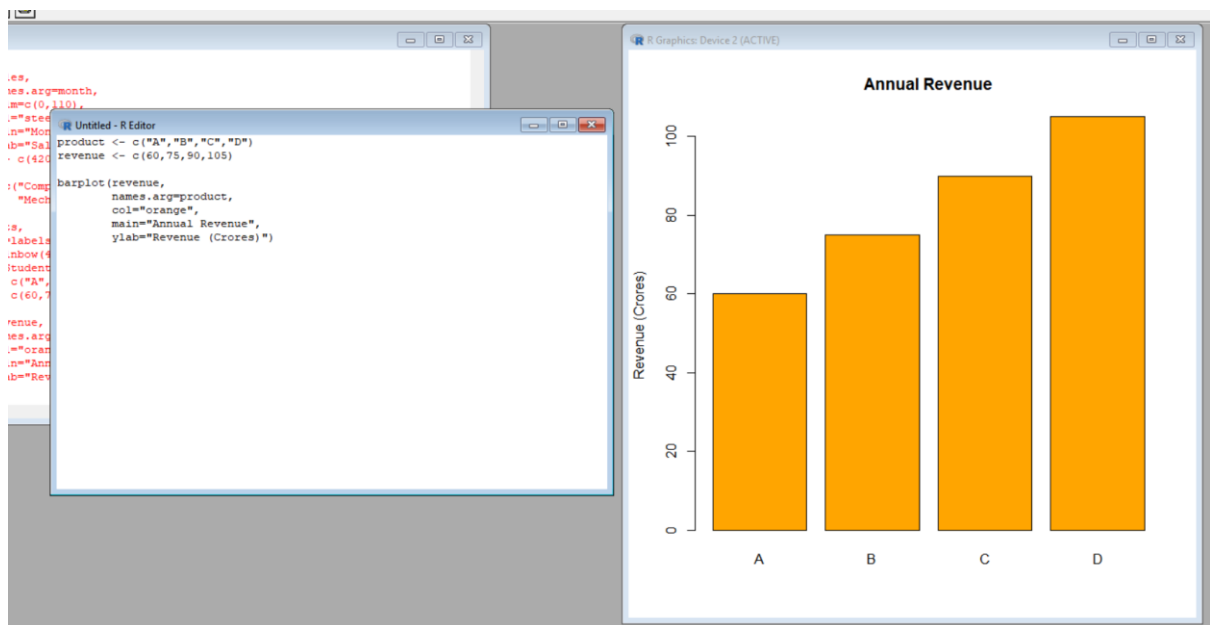
R code:

```
product <- c("A","B","C","D")
```

```
revenue <- c(60,75,90,105)
```

```
barplot(revenue,  
        names.arg=product,  
        col="orange",  
        main="Annual Revenue",  
        ylab="Revenue (Crores)")
```

Output:



Q4) i. Why are 3D charts misleading?

Perspective changes the apparent height and width of bars.

Objects closer to the viewer appear larger than those farther away.

Therefore, the visual difference becomes much larger than the real numerical difference.

ii. **Actual vs Perceived Difference**

Actual Difference

$$550 - 500 = \mathbf{50 \text{ GWh}}$$

Percentage Difference

$$50/500 \times 100 = \mathbf{10\%}$$

However,

the 3D chart makes Hydro appear **40% larger** than Wind.

Thus, the perceived difference is four times the actual difference.

iii. **Better Visualization**

Use

- Simple 2D bar chart
- Horizontal bar chart

These accurately represent the values.

iv. **Importance**

Accurate sustainability reporting is essential because governments, investors, and environmental agencies depend on these reports for planning renewable energy policies and investments.

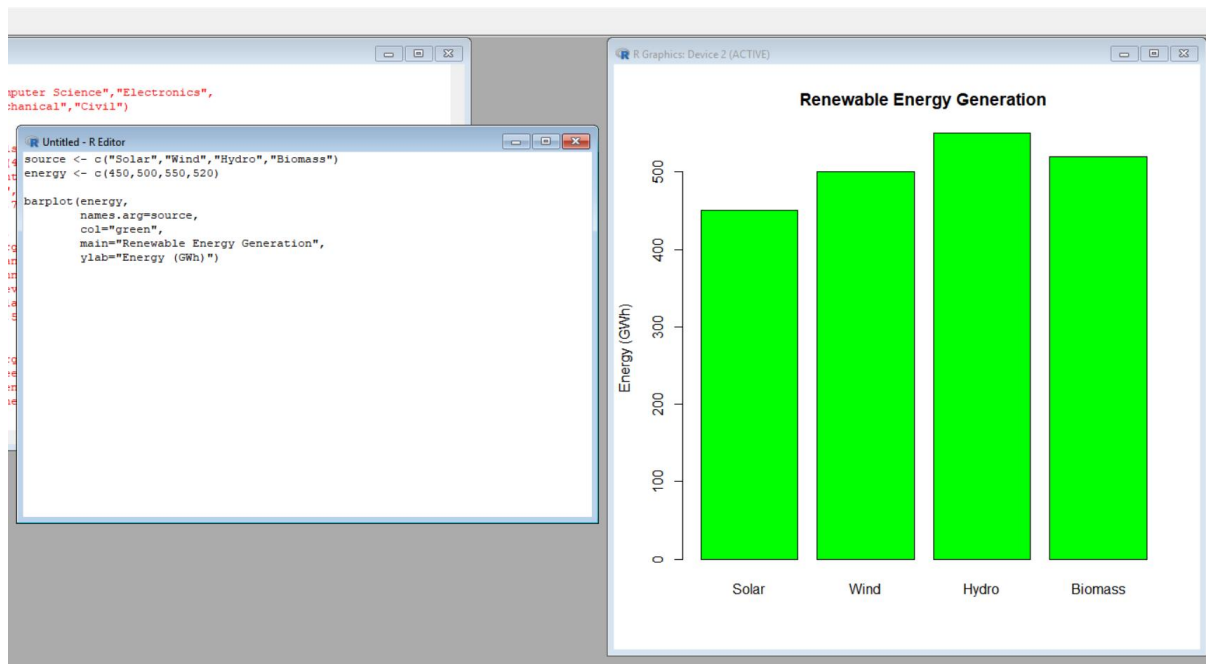
R code:

```
source <- c("Solar","Wind","Hydro","Biomass")
```

```
energy <- c(450,500,550,520)
```

```
barplot(energy,  
        names.arg=source,  
        col="green",  
        main="Renewable Energy Generation",  
        ylab="Energy (GWh)")
```

Output:



Q5) i. Why is radius scaling misleading?

The size of a bubble is determined by its **area**, not its radius.

If the radius is directly proportional to the data,

$$\text{Area} = \pi r^2$$

This causes the area to increase much faster than the actual values.

As a result, District E appears much larger than it should.

ii. Correct Scaling Method

The **area** should be proportional to the data.

Therefore,

Radius

$$= \sqrt{\text{Value}}$$

or

Radius

$$= \sqrt{\text{Value}/\pi}$$

This ensures that the bubble area correctly represents the data values.

iii. Alternative Visualization

A bar chart is the best alternative because it enables accurate comparisons.

iv. Public Health Impact

An incorrect bubble chart may

- Allocate too many medical resources to one district.
- Reduce supplies for other districts.
- Mislead policymakers.
- Affect emergency healthcare planning.

R code:

```
district <- c("A","B","C","D","E")
```

```
cases <- c(150,250,350,450,550)
```

```
symbols(1:5,  
        rep(1,5),  
        circles=sqrt(cases),  
        inches=0.5,  
        bg="red",  
        xlab="District",  
        ylab="")
```

```
axis(1,at=1:5,labels=district)
```

```
title("Disease Cases by District")
```

Output:

```
range",
Annual Revenue",
Revenue (Crores) ")
plot", "Wind", "Hydro", "Biomass")
}, 5
# Untitled - R Editor
district <- c("A", "B", "C", "D", "E")
cases <- c(150, 250, 350, 450, 550)
symbols(1:5,
rep(1, 5),
circles=sqrt(cases),
inches=0.5,
bg="red",
xlab="District",
ylab="")
axis(1, at=1:5, labels=district)
title("Disease Cases by District")
lab
Ca
```

R Graphics: Device 2 (ACTIVE)

Disease Cases by District

